

Question				Marking details	Marks available														
					AO1	AO2	AO3	Total	Maths	Prac									
3	(a)			The alpha particle / it has <u>two</u> fewer [accept: less] electrons / no electrons [Not: just less / fewer electrons] [Treat 'it' as the alpha particle] Reference to charge neutral – not credited or penalised	1			1											
	(b)	(i)		<table border="1"><thead><tr><th></th><th>Element</th><th>Nucleon number</th></tr></thead><tbody><tr><td>X</td><td>Pb / lead (1)</td><td>214 (1)</td></tr><tr><td>Y</td><td>Bi / bismuth (1)</td><td>214 (1)</td></tr></tbody></table> NB. Additional numbers in the nucleon number box → no mark.		Element	Nucleon number	X	Pb / lead (1)	214 (1)	Y	Bi / bismuth (1)	214 (1)		4		4		
	Element	Nucleon number																	
X	Pb / lead (1)	214 (1)																	
Y	Bi / bismuth (1)	214 (1)																	
		(ii)		Z		1		1											
				Question 3 total	1	5	0	6	0	0									